Proxy Master — Joint Ring and Shortcut Search (MILP)

Ring topology and abstract shortcut placement in one model: the lower-bound stage of the pipeline

Goal. Search over ring topologies and abstract shortcut placements simultaneously, and return a valid lower bound on the worst-case demand path length together with a pool of promising candidate rings. The master is deliberately a relaxation: it never constructs real shortcut geometry.

1 Setup

N nodes with fixed coordinates (x_i, y_i), index set $V = \{0, \dots, N-1\}$. All waveguides are axis-parallel, so distances are Manhattan, $\delta(i,j) = |x_i - x_j| + |y_i - y_j|$. A ring is a Hamiltonian cycle; each ring edge is drawn as an L-shape (horizontal-then-vertical HV, or vertical-then-horizontal VH) and no two ring waveguides may cross. A demand $q \in Q$ sends one unit of signal from source $s(q)$ to sink $t(q)$; the demand set Q is given by the demand matrix.

Shortcut candidates. $\mathcal{P}$ is the set of abstract shortcut pairs, built once from the node coordinates. Each $c \in \mathcal{P}$ carries *only* its endpoints $i(c)$, $j(c)$ and the abstract length $\delta(c) = \delta\big(i(c), j(c)\big)$. No routed path is associated with c in the master.

Intuition

The master answers "which ring shape, together with which pairs of nodes bridged, could in the best case give a short worst path?" It is allowed to be optimistic about the bridges, because a bridge can never be shorter than the straight Manhattan distance between its two endpoints. That optimism is exactly what makes the answer a lower bound.

2 Variables

$b_{ij} \in \{0,1\}$	1 if the ring goes directly $i \to j$
$r_{ij} \in \{0,1\}$	L-shape of an active ring edge (0 = HV, 1 = VH)
$z_c \in \{0,1\}$	1 if the abstract shortcut pair $c \in \mathcal{P}$ is selected
$f^q_{ij} \in [0,1]$	proxy flow of demand q over the ring movement $i \to j$
$f^q_{c,d} \in [0,1]$	proxy flow of demand q through shortcut c in direction $d \in \{0,1\}$
$W \geq 0$	worst-case proxy path length (the objective)
$u_i \in [1, N-1]$	MTZ order variables, small N only

Direction $d = 0$ means $i(c) \to j(c)$, direction $d = 1$ means $j(c) \to i(c)$. The flow variables are continuous; they only serve to bound path lengths.

3 Ring constraints

"The active edges form one crossing-free Hamiltonian ring."

(M1)–(M2) One successor, one predecessor.

$$\sum_{j \neq i} b_{ij} = 1 \quad \forall i, \qquad\qquad \sum_{i \neq j} b_{ij} = 1 \quad \forall j. \tag{1}$$

(M3) No two-cycles.

$$b_{ij} + b_{ji} \leq 1 \qquad \forall \{i,j\}. \tag{2}$$

(M4) No sub-tours. Small $N \leq 20$: Miller–Tucker–Zemlin (MTZ),

$$u_i - u_j + N\, b_{ij} \leq N - 1 \qquad \forall\, i, j \neq 0. \tag{3}$$

Large N: drop MTZ and add a lazy cut whenever a small loop S appears in an incumbent, $\sum_{(i,j) \in S} b_{ij} \leq |S| - 1$.

(M5) Orientation activation. Orientation only matters on arcs that are actually selected:

$$r_{ij} \leq b_{ij} \qquad \forall\, (i,j). \tag{4}$$

(M6) Crossing-free ring geometry. For every geometric conflict — edge (i,j) with shape o_1 and edge (k,ℓ) with shape o_2 whose L-segments intersect — forbid selecting both with those shapes:

$$b_{ij} + b_{k\ell} + [\, r_{ij} = o_1\,] + [\, r_{k\ell} = o_2\,] \leq 3. \tag{5}$$

If both edges are active and both use the crossing shapes the left side is 4 — forbidden, so one edge must flip its L-shape or be dropped. A companion family forbids an L-shape routed through a third node (pass-through triples). **(M6)** is what separates this ring from an ordinary shortest tour: the tour must also be drawable with non-overlapping L-shaped waveguides.

4 Shortcut constraints

(M7) Node incidence limit. A node may carry at most one shortcut (hardware / power-distribution rule):

$$\sum_{c\,:\, v \in \{i(c), j(c)\}} z_c \leq 1 \qquad \forall v \in V. \tag{6}$$

Selected shortcuts therefore form a matching on the nodes, at most $\lfloor N/2 \rfloor$ of them.

(M8) No shortcut duplicating a ring edge. A shortcut between two ring neighbours is meaningless:

$$z_c + b_{i(c)j(c)} + b_{j(c)i(c)} \le 1 \qquad \forall c \in \mathcal{P}. \tag{7}$$

(M9) Always-crossing pairs, cardinality form. Let $\mathcal{X}(c) \subseteq \mathcal{P}$ be the partners of c for which *all four* combinations of minimal L-shapes (VH/HV for each) intersect. Such a pair can only be realised as a crossing, so it consumes crossing budget:

$$\sum_{p \in \mathcal{X}(c)} z_p \;\le\; 1 + |\mathcal{X}(c)|\,(1 - z_c) \qquad \forall c \in \mathcal{P}. \tag{8}$$

If $z_c = 1$ the right-hand side is 1: at most *one* always-crossing partner may be selected alongside c. If $z_c = 0$ the constraint is vacuous. This is the same physical rule the Shortcut Solver enforces on real routed paths — at most one crossing per shortcut — so the screen stays inside what a realisable solution may do.

5 Proxy routing

(M10) Ring flow activation.

$$f^q_{ij} \le b_{ij} + b_{ji} \qquad \forall q,\ \forall (i,j). \tag{9}$$

Flow may traverse a selected undirected ring edge in either direction. The tour direction encoded by b constrains the structure, not the routing — this matches the physical model of two independent directional waveguides (CW and CCW) on the ring.

(M11) Shortcut flow activation. No capacity limit is imposed: any number of demands, in either direction, may share a selected shortcut.

$$f^q_{c,d} \le z_c \qquad \forall q,\ \forall c \in \mathcal{P},\ \forall d \in \{0,1\}. \tag{10}$$

(M12) Flow conservation. For every demand q and node v, with $\Delta^+_{\mathrm{sc}}(v)$ the shortcut hops leaving v and $\Delta^-_{\mathrm{sc}}(v)$ those entering v:

$$\sum_{j \ne v} f^q_{vj} - \sum_{j \ne v} f^q_{jv} + \sum_{(c,d) \in \Delta^+_{\mathrm{sc}}(v)} f^q_{c,d} - \sum_{(c,d) \in \Delta^-_{\mathrm{sc}}(v)} f^q_{c,d} = \begin{cases} +1 & v = s(q) \\ -1 & v = t(q) \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

(M13) Worst-case bounding.

$$W \;\ge\; \sum_{(i,j)} \delta(i,j)\, f^q_{ij} \;+\; \sum_{c \in \mathcal{P}} \sum_{d \in \{0,1\}} \delta(c)\, f^q_{c,d} \qquad \forall q \in Q. \tag{12}$$

6 Objective

$$\min\ W. \tag{13}$$

Nothing else: no ε-regulariser, no bend term, no wavelength term. The model is a min–max multicommodity proxy.

7 No-good cuts

After the outer loop has evaluated a ring R with the Shortcut Solver, that ring is excluded from all later master solves. Because b is directed, two constraints are needed, one per traversal direction. Let E_R be the directed tour arcs of the evaluated ring:

$$\sum_{(i,j) \in E_R} b_{ij} \le N - 1, \qquad \sum_{(i,j) \in E_R} b_{ji} \le N - 1. \tag{14}$$

Intuition

A Hamiltonian tour sets exactly N of the directed arc variables — one per edge. Each cut therefore forces at least one arc of the excluded tour to flip. The forward cut removes the tour as found, the reverse cut removes the same cycle traversed backwards, which is a different assignment of b but the same geometric ring.

8 Why the master is a lower bound

Two independent relaxations point in the same direction.

1. **Optimistic shortcut length.** Any realised shortcut between $i(c)$ and $j(c)$ has routed length $L_c \ge \delta(c)$, since obstacles can only lengthen a Manhattan path. The master charges $\delta(c)$.
2. **Missing geometry.** The master imposes no ring $\leftrightarrow$ shortcut interaction beyond the screens above and does not require a shortcut to be routable at all. Its feasible set therefore contains every realisable configuration.

Hence for every ring R,

$$LB(R) \;\le\; W_{\mathrm{true}}(R), \tag{15}$$

and the master optimum LB_k is a lower bound over all rings not yet cut. Since cuts only remove solutions, LB_k is non-decreasing across rounds — the property the termination proof rests on.

9 Solution pool

A min–max objective admits large plateaus: many structurally different rings share the same LB. The master is therefore solved with a proven k-best pool (`PoolSearchMode=2`, `PoolGap=0`) and the returned solutions are filtered to those with objective $= LB_k$. One solve then yields several tied optimal rings, which the outer loop can rank and verify without re-solving in between.

Notes

- **Metric.** All lengths are plain Manhattan millimetres in W, matching the Shortcut Solver and the shared worst-case metric. Insertion-loss terms $\alpha \cdot (\text{length}) + \beta \cdot (\#\text{edges})$ are reporting-only and never enter the bound.
- **Incremental session.** One persistent Gurobi model keeps the variables and structural constraints, appends the new cuts across rounds and re-optimises. This is operationally cheaper but mathematically identical to rebuilding with all cuts present.